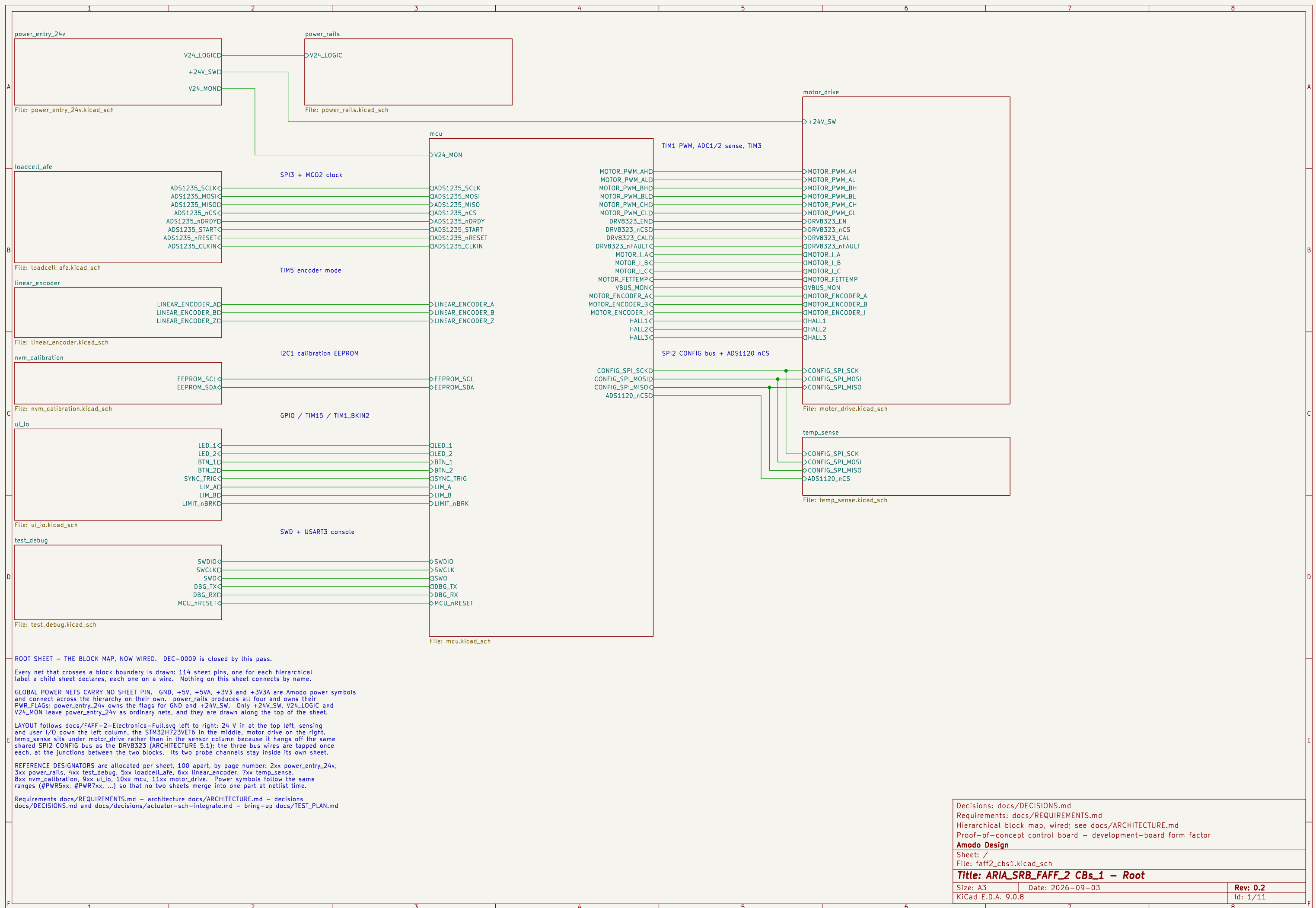
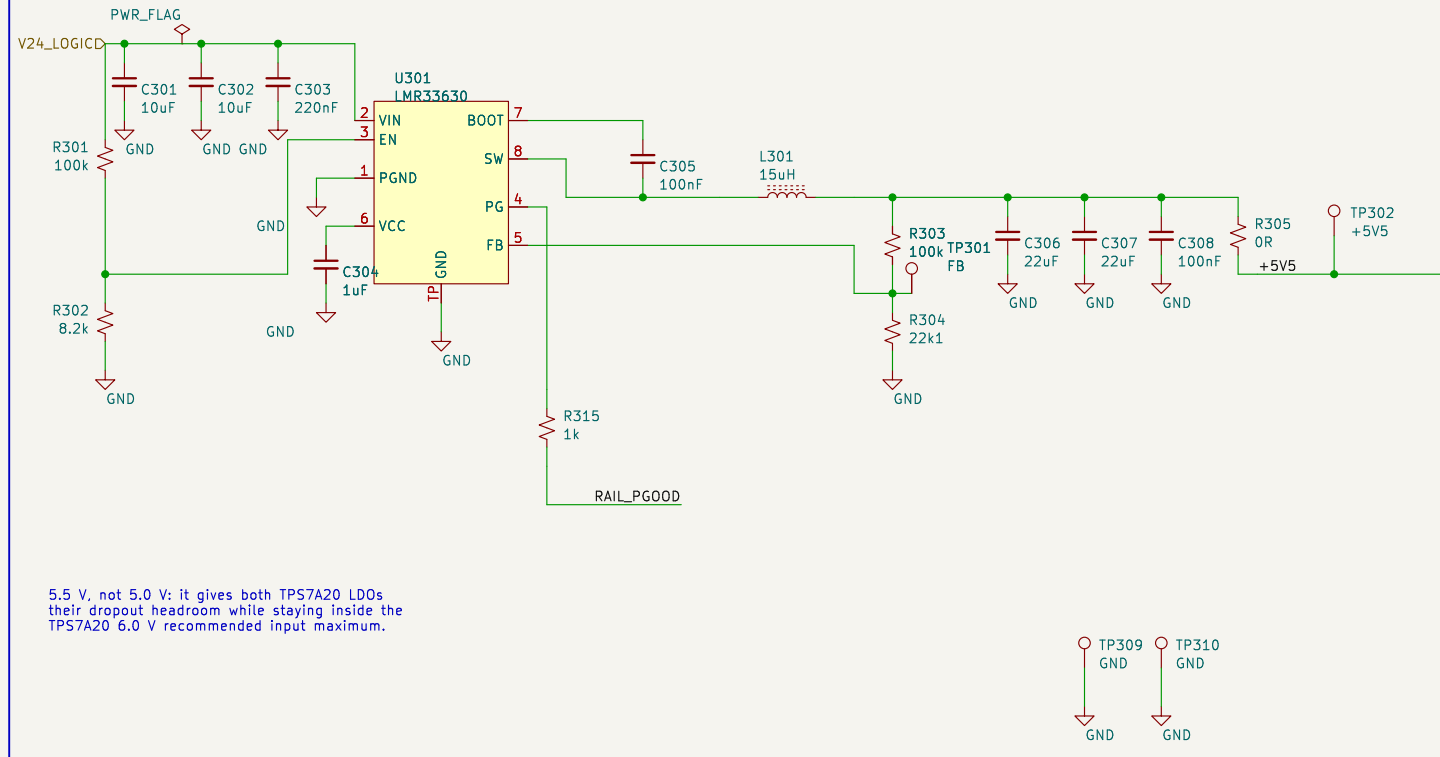
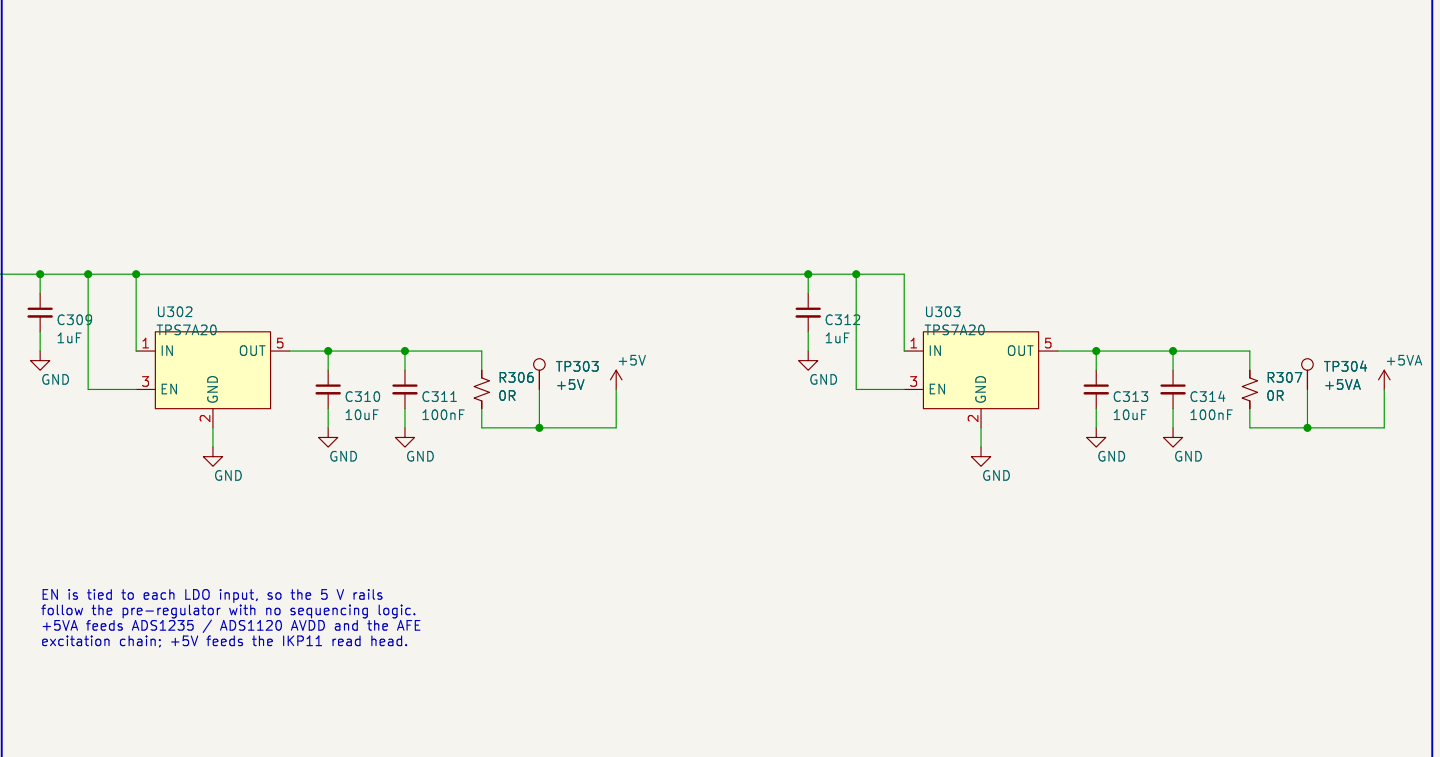




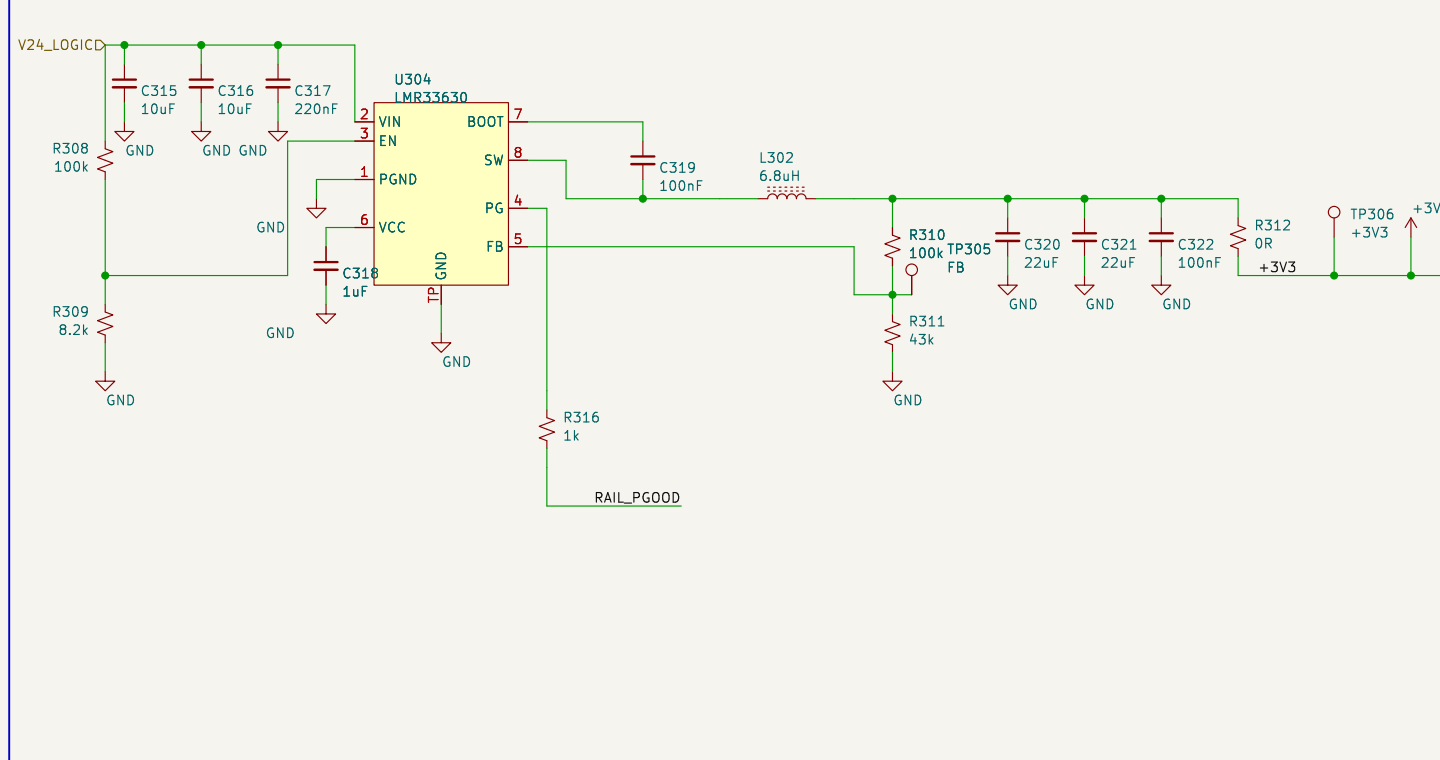
A. PRE-REGULATOR V24_LOGIC -> 5.5 V (LMR33630, 400 kHz)



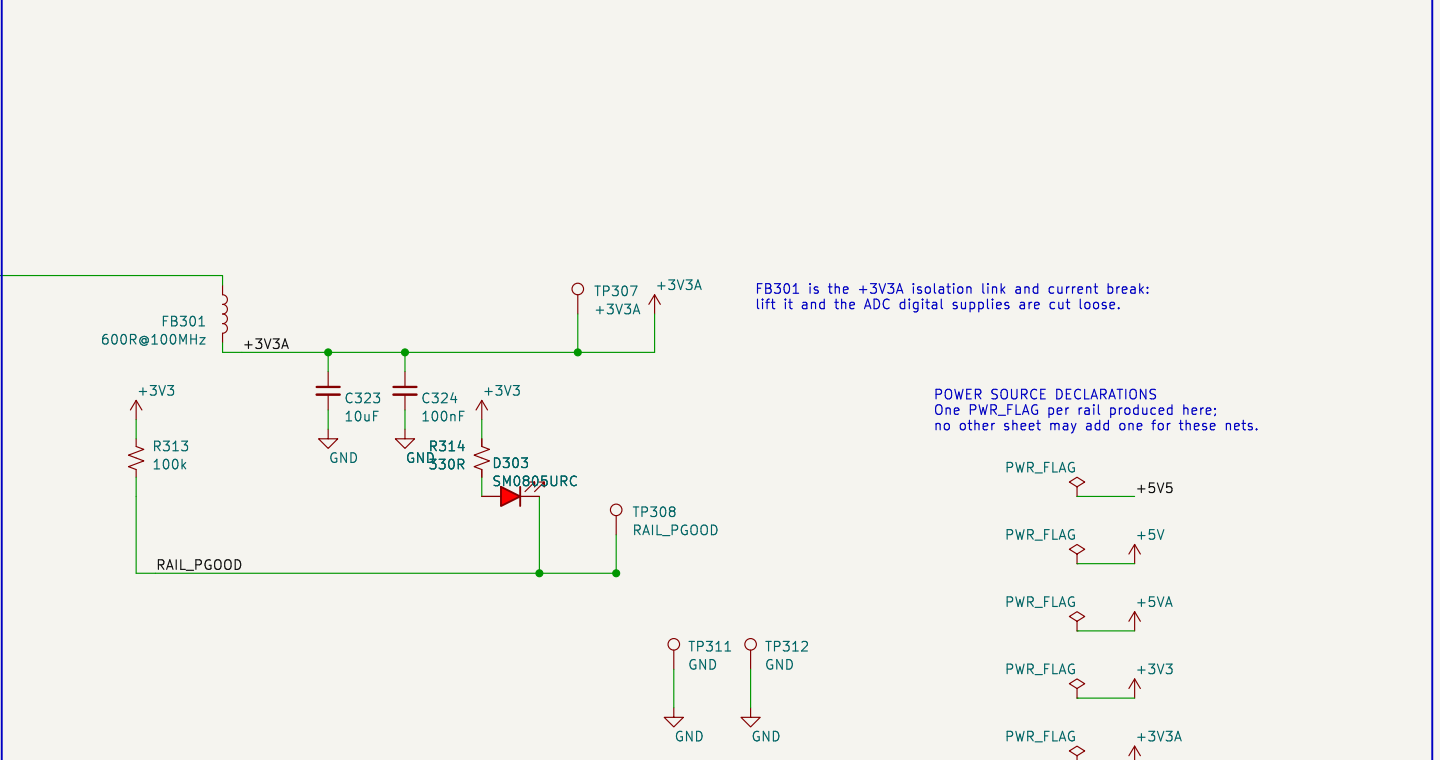
B. POST-REGULATED 5 V RAILS (TPS7A20: +5V digital, +5VA analog)



C. 3V3 REGULATOR V24_LOGIC -> 3.3 V (LMR33630, 400 kHz)



D. ANALOG 3V3 FOR THE ADC DIGITAL SUPPLIES, AND RAIL POWER-GOOD



BLOCK: power_rails – rail set and topology; answers OQ-02 and OQ-07

RAILS	+5V5	LMR33630 buck from V24_LOGIC. Pre-regulator only. local to this sheet.
	+5V	TPS7A20 LDO from +5V5. IKP11 read head (<65 mA plus RS-422 drive) and spare.
	+5VA	TPS7A20 LDO from +5V5. ADS1235 / ADS1210 AVDD and the AFE 5 V excitation and reference chain, which lives in loadcellAfe - take it from +5VA, not +5V.
	+3V3	LMR33630 buck from V24_LOGIC. ST1832, +5SB3320, DRV8323 logic, QSPI, EEPROM, RS-422 receiver, rotary encoder. -1.1 A worst case against a 3 A part.
	+3V3SA	ferrite-isolated from +3V3 for the ADC digital supplies.

WHY A PRE-REGULATOR The analog 5 V must be regulated separately from the digital 5 V or the read head switching current lands on the ADC supply, and an LDO needs headroom above 5 V to do it. One buck at 5.5 V feeding two LDOs costs one converter and post-regulates both 5 V rails. 400 kHz on both bucks: 3.3 V at 1.4 MHz would need a ~98 ns on-time, too close to the minimum.

PER RAIL (docs/TEST_PLAN.md 3.1): one OR link that is BOTH the isolation link and the current-measurement break, a test point downstream of it, and a test point on each feedback node: R305 +5V5, R306 +5V, R307 +5VA, R312 +3V3, FB301 +3V3A. Per-CONSUMER links are not here: every block takes the same global rail net, so per-block breaks would need per-block rail names and would break the frozen block contracts. A block that wants one fits a OR at its own supply entry, inside its sheet.

SEQUENCING None required. The shared EN divider (16.2 V rising) holds both converters off until the 24 V input is real; the LDOs follow +5V5 and +3V3A follows +3V3.

INTERFACES IN V24_LOGIC <- power_entry_24v (owns its PWR_FLAG).
RAILPGOOD is open drain, the wired-AND of both converters
through R315 / R316. It stays SHEET-LOCAL: LED D303 and TP308
are its only consumers until OQ-07 allocates an MCU GPIO.
Global power nets produced here: +5V, +5VA, +3V3, +3V3SA.
GND is board-wide, one net; power_entry_24v owns its PWR_FLAG.

Reasoning, part sizing and residual ERC:
docs/decisions/actuator-sch-power.md

Design reasoning: docs/decisions/actuator-sch-power.md

Rail set and topology answer OQ-02 / OQ-07

Per-rail isolation link / current break and test point

24 V $\rightarrow$ 5V5 pre-reg $\rightarrow$ +5V and +5VA; 24 V $\rightarrow$ +3V3 $\rightarrow$ +3V3A

Amodo Design

Sheet: /power_rails/

File: power_rails.kicad_sch

Title: CBs_1 – Power Rails

Size: A3

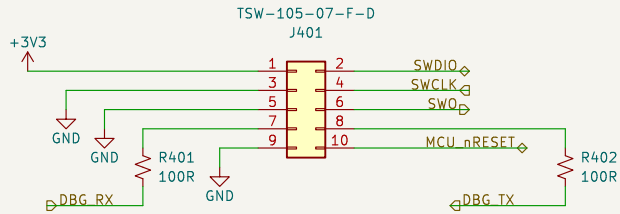
Date: 2026-09-02

8

Rev: 0.2

Id: 3/11

T1 DEBUG HEADER – SWD + USART3 CONSOLE (REQ-AR-15)

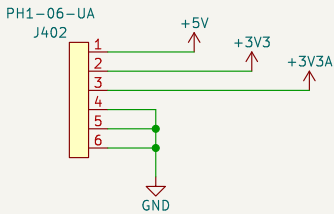


PINOUT – standard ARM Cortex Debug 10-pin order on 2.54 mm pitch, so a 2x5 SWD probe cable plugs straight on:
1 VTref (+3V3) 2 SWDIO 3 GND 4 SWCLK 5 GND
6 SWO 7 DBG_RX 8 DBG_TX 9 GND 10 nRESET

Pins 7 and 8 are KEY / NC on the ARM pinout and are reused here for the USART3 console (REQ-AR-15 and the block diagram both put SWD and UART on ONE header). They are SWD-safe; an SWD probe drives only 2, 4 and 10 and reads 6. This part is SWD-only anyway – the .ioc sets Trace_Aynchronous_SW, so PB4/PB3 are not JTAG. The 100R series resistors make a mis-plugged JTAG probe or a shorted console pin a nuisance rather than damage.

DBG_TX / DBG_RX keep the .ioc net names: DBG_TX is the STM32 OUTPUT (USART3_TX, PD8), DBG_RX the STM32 INPUT (USART3_RX, PD9). nRESET is MCU_nRESET, generated in the mcu block (which owns its 100 nF, its push button and its test point – producer-owns-the-break, DEC-0007).

T2 CONSOLIDATED RAIL PROBE HEADER (TEST_PLAN 4)



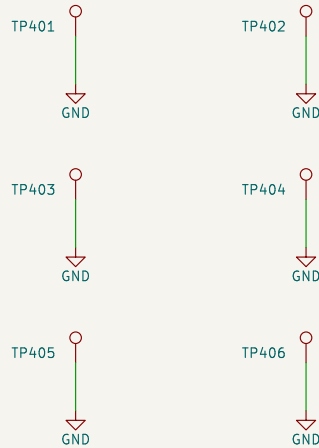
One header to check every logic rail with a DMM during bring-up steps 2 and 3 of TEST_PLAN 5. The rails themselves, and their isolation links and current breaks, belong to power_rails – this header only observes them.

1 +5V 2 +3V3 3 +3V3A 4 GND 5 GND 6 GND

+24V IS DELIBERATELY NOT ON THIS HEADER. 24 V next to 3.3 V on a 2.54 mm strip is one slip away from destroying the board; power_entry_24v already carries the +24V_SW test point and current break that TEST_PLAN 4 asks for.

+3V3A is the MCU VDDA / VREF+ analogue rail the mcu stub note lists as an input. If power_rails names that rail differently, this header and the mcu sheet both follow it – see docs/decisions/actuator-sch-mcu.md.

T3 GND HOOKS FOR SCOPE CLIPS



Six through-hole GND loops, spread across the board so an oscilloscope ground clip is never far from the signal hook being probed. The house standard asks for these wherever TestPointHook is used, and the block sheets use hooks on I2C1, SPI2, SPI3 and the fault lines.

SHEET NOTES

BLOCK: test_debug – system-level bring-up hardware that no functional block owns (DEC-0007). Everything here observes; nothing here generates a rail or a signal, so this sheet owns no isolation link or current break. Per-block test points, links and breaks stay in the sheet that produces the net.

ROOT SHEET NEEDS – hierarchical labels this sheet exposes, all of them shared with the mcu block: SWDIO, SWCLK, SWO, MCU_nRESET, DBG_TX, DBG_RX. The rails (+5V, +3V3, +3V3A, GND) are global power nets and need no sheet pins. All six are sheet pins on the test_debug block of the root sheet now, wired straight across to the mcu block: the integration pass that closed DEC-0009 drew them, and project ERC is clean at severity-all. See docs/decisions/actuator-sch-mcu.md for the full list and directions.

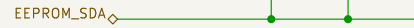
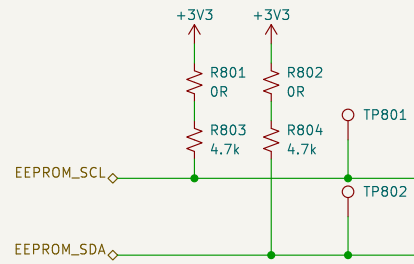
Reasoning: docs/decisions/actuator-sch-mcu.md
Bring-up order and gates: docs/TEST_PLAN.md
Per-block test points live in their own sheets (DEC-0007)
One debug header carries SWD and USART3 (REQ-AR-15)

Amodo Design

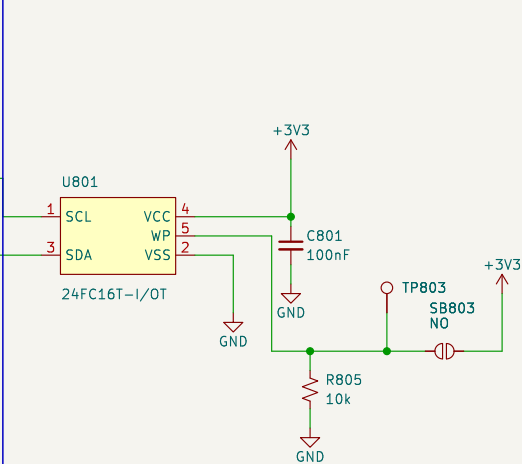
Sheet: /test_debug/
File: test_debug.kicad_sch

Title: CBs_1 – Test, Debug and Bring-Up

Size: A3 Date: 2026-09-02 Rev: 0.2
KiCad E.D.A. 9.0.8 Id: 4/11

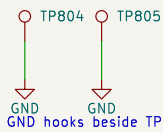


R801/R802 are the removable links of TEST_PLAN 4: lift either to drop its pull-up. SB801/SB802 are cut to take the EEPROM off I2C1 while the bus stays pulled up – this block owns that break, so mcu must not duplicate it. The SDA pull-up crosses the SCL run, no junction.



WP is ACTIVE HIGH. R805 holds it low, so the part is writable and firmware can store calibration at any time; solder SB803 to lock the device permanently. TP3 lets a jig drive WP either way. This resolves OQ-06 – no MCU pin is spent on write protect.

GROUND HOOKS



+3V3 and GND are generated in power_rails and enter this block through the hierarchy; nothing here drives them, so ERC reports power_pin_not_driven until that sheet lands. See docs/decisions/actuator-sch-periph.md.

analyse clips.

BLOCK NOTES – nvm_calibration

U1 = Microchip 24FC16T-I/OT, 16 kbit (2048 x 8) I2C EEPROM in SOT-23-5.

Pin map verified against the datasheet package drawing (DS20001703R p.1):

1 SCL 2 VSS 3 SDA 4 VCC 5 WP.
1 7-5.5 V and 1 MHz capable, so it keeps

1.7–5.5 V and 1 MHz capable, so it keeps up with I2C1 at either 400 kHz or Fast-mode Plus, and 4 million erase/write cycles covers repeated calibration.

2 KB is sized for coefficients, not tables: force and temperature calibration constants, the variant (V50N / V10N – see DEC-0010, where the variants differ only by a calibration constant), board serial number and build state. Bulk force profiles live in the OCTOSP11 RAM on the mcu sheet (DEC-0006), not here.

ADDRESSING CAVEAT. The 24xx16 uses all three block-select bits internally, so it answers to the **WHOLE** 1010xxx address space and A0/A1/A2 do not exist on the part. No second 1010-family device can share I2C1. It is the only device on the bus today; if another is ever added, swap to a smaller 24xx02/04 first.

4k7 pull-ups suit 400 kHz on a short board-level bus with one load. They are the only pull-ups on I2C1: the mcu block must not add its own.

Test strategy: docs/TEST_PLAN.md
REQ-EL-08, REQ-AR-11; resolves OQ-06 (write protect)
I2C1: EEPROM_SCL PB8, EEPROM_SDA PB9
16 kbit I2C EEPROM for calibration and compensation data

Amodo Design

Sheet: /nvm_calibration/
File: nvm_calibration.kicad_sch

Title: CBs_1 – NVM / Calibration Store

Size: A3

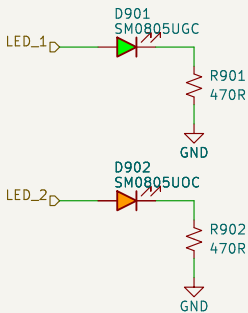
Date: 2026-09-02

Rev: 0.2

KiCad E.D.A. 9.0.8

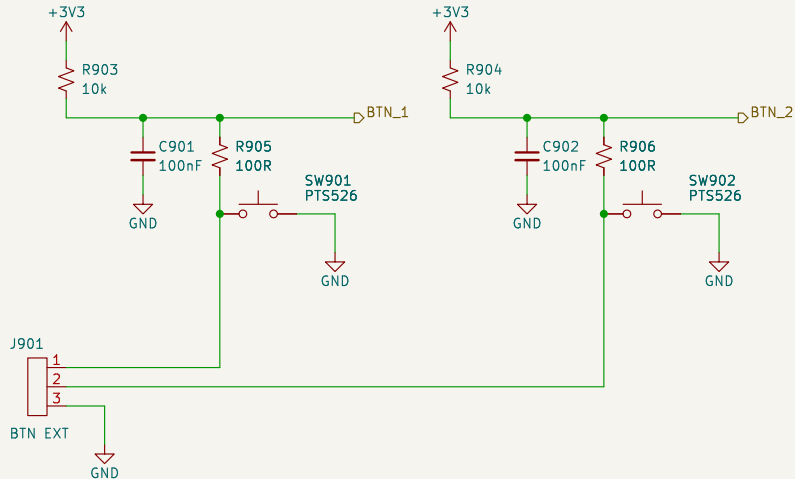
Id: 8/11

STATUS LEDs



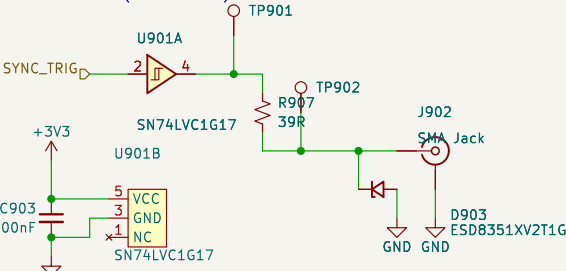
Active high, about 2.8 mA each.
LED_1 green is the heartbeat of
TEST_PLAN bring-up step 3; LED_2
orange is the fault / status lamp.

USER BUTTONS (BTN_1 = HOME)



REQ-CC-06 asks for homing from the API OR an external button, so J901 parallels both on-board switches for a panel button. R905/R906 limit the contact current and set the debounce with C901/C902 – about 1 ms rise, 10 us fall. Which of BTN_1 / BTN_2 is HOME is OQ-05: firmware and silkscreen only, the two channels are electrically identical.

SYNC / TRIGGER OUT (TIM15_CH1)



REQ-EL-05 / REQ-EL-06: 3.3 V push-pull, 50 ohm source terminated, SMA jack. U901 buffers PE5 so the MCU pin never sees the cable and the source impedance is repeatable: R907 39 R plus the LVC output impedance (10–15 ohm typical) is about 50 ohm. TP901 is pre-termination, TP902 post; the SMA itself is the coaxial access of TEST_PLAN 2.3. D903 is 0.55 pF and does not soften the edge.

BLOCK NOTES – ui_io

SAFETY. SPEC 9.1 asks only for a firmware kill on limit actuation (REQ-SF-02). The block diagram and the .ioc both carry a second, independent hardware path (REQ-SF-05, DEC-0012): U902 ANDs the two healthy-limit nodes, so LIMIT_nBRK is HIGH only while both switches are closed and both harness legs are intact. Any limit reached, any broken wire, an unplugged J-LIM or a dead 3V3 rail drives it LOW into TIM1_BKIN2, and the PWM outputs go inactive with firmware taking no part. LIM_A / LIM_B reach the MCU on their own EXTI pins for the firmware kill and for telling the two ends apart. The AND gate's third input is tied high.

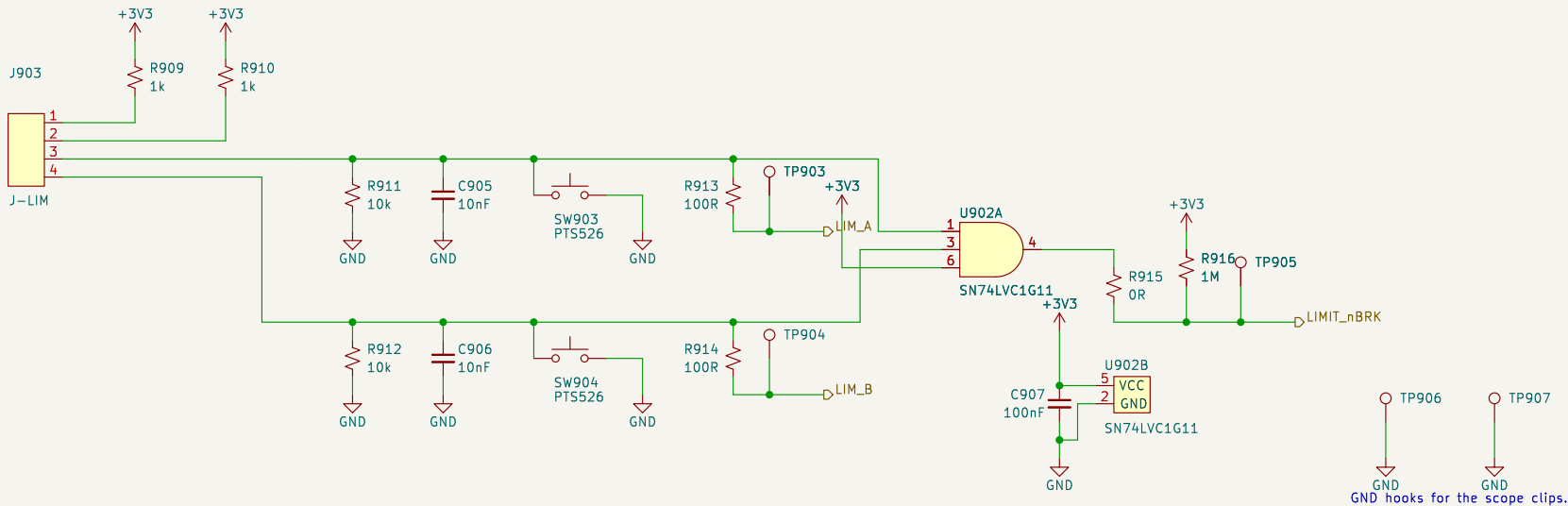
R915 is the TEST_PLAN 3.3 link that lets the two TIM1 BREAK sources be exercised alone. It is fitted by default and must be marked on the silkscreen. With it lifted, R916 (1 M) holds LIMIT_nBRK high so the DRV8323_nFAULT path can be tested on its own; the firmware kill through LIM_A / LIM_B is unaffected either way. 1 M is weak enough that a tripped chain still pulls the net down to about 30 mV.

SW903 / SW904 are the 'assert each limit without the mechanics' means of TEST_PLAN 4: pressing one pulls its node low through the 1 k feed (3.3 mA), which is exactly the state that end of stroke produces. They act independently because the two switch loops are independent – a series safety chain could not be stimulated one end at a time.

C905 / C906 (10 nF) with the -0.9 k source give about 9 us of rise filtering and 100 us of fall; at the 20 mm/s of REQ-ME-03 that is 2 um of extra travel before the break asserts, against millimetres of switch overtravel. R913 / R914 protect the MCU pins. No discrete TVS is fitted on the limit lines: the harness stays inside the instrument and the RC plus the series resistor already bound the current into the MCU clamps. The SMA is the one externally exposed pin here, and it does get a clamp (D903).

+3V3 and GND are generated in power_rails and enter this block through the hierarchy; nothing here drives them, so ERC reports power_pin_not_driven until that sheet lands.

END-OF-STROKE LIMIT SWITCHES + TIM1 BREAK



J-LIM pinout: 1 LIM_A feed 2 LIM_B feed 3 LIM_A return 4 LIM_B return.
Both switches are NORMALLY CLOSED, so a closed switch pulls its node HIGH (3.0 V) and an open switch – or a broken wire, or an unplugged connector – lets the 10 k pull-down take it LOW. LOW therefore means 'at that end of stroke, or the wiring has failed', which is the safe direction to fail in.

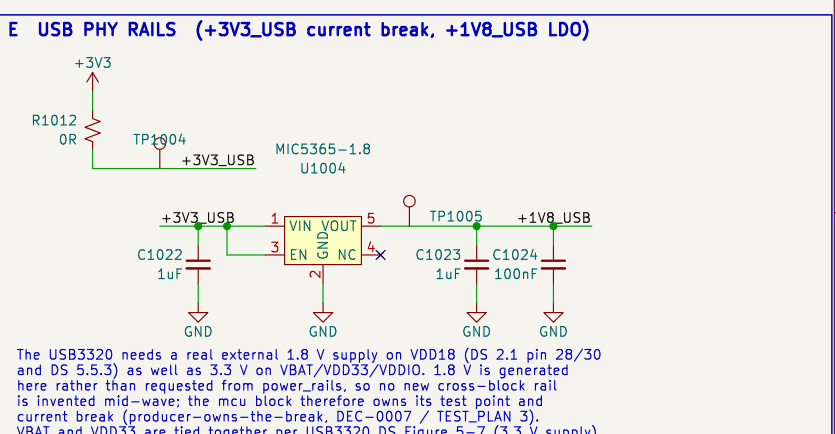
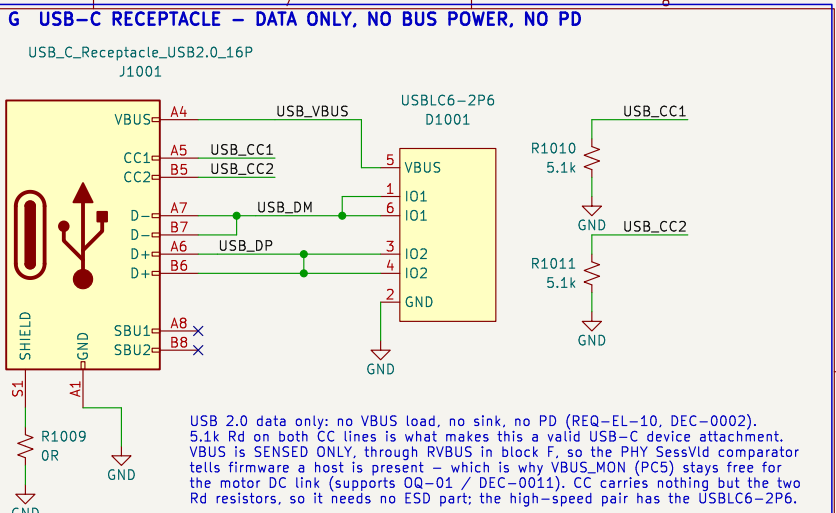
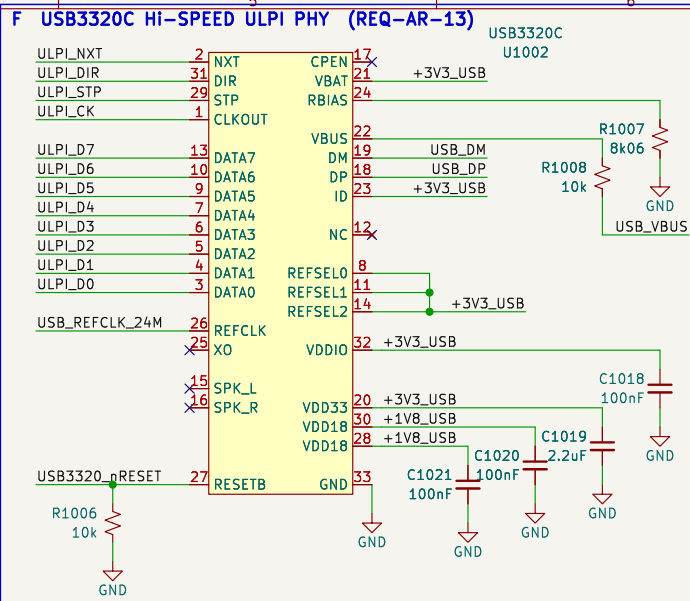
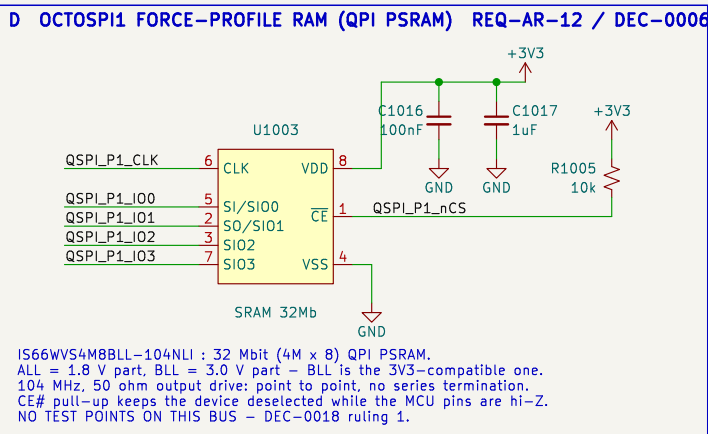
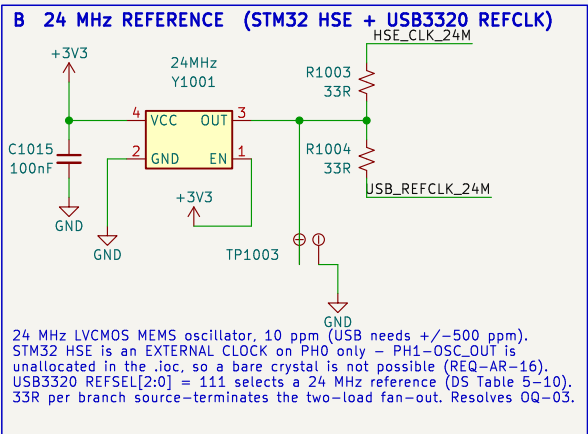
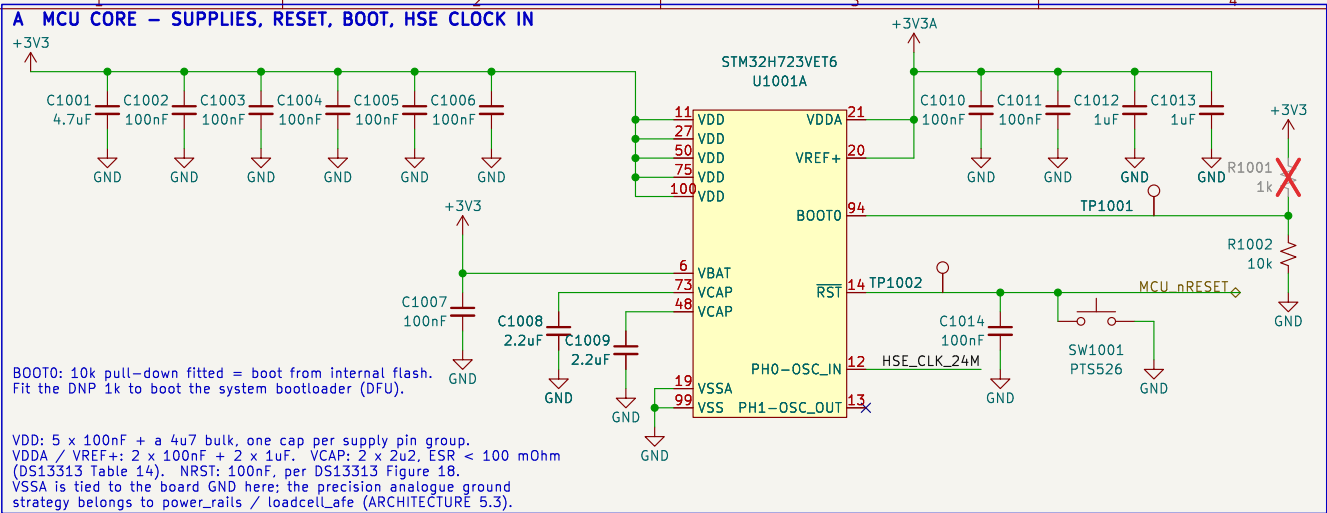
Test strategy: docs/TEST_PLAN.md
REQ-EL-05/06/09, REQ-CC-06, REQ-SF-01/02/05
LIMIT_nBRK PE6 -> TIM1_BKIN2: hardware motor kill (DEC-0012)
Status LEDs, user buttons, SMA SYNC output, end-of-stroke limits

Amodo Design

Sheet: /ui_io/
File: ui_io.kicad_sch

Title: CBs_1 – User I/O, SYNC, Limit Switches

Size: A3 Date: 2026-09-02 Rev: 0.2
KiCad E.D.A. 9.0.8 Id: 9/11



MCU PORTS – PIN ASSIGNMENT IS THE .ioc (DEC-0013); ULPI / QSPI / USB3320_nRESET / SPARE_* stay on this sheet

LINEAR_ENCODER_A	22	PA0
LINEAR_ENCODER_B	23	PA1
LINEAR_ENCODER_Z	24	PA2
ULPI_D0	25	PA3
MOTOR_FETTEMP	26	PA4
ULPI_CK	29	PA5
MOTOR_I_A	30	PA6
MOTOR_I_C	33	PA7
SPARE_PA8	67	PA8
CONFIG_SPI_SCK	68	PA9
SPARE_PA10	69	PA10
LED_2	70	PA11
LED_1	71	PA12
SWDIO	72	PA13_JTMS/SWDIO
SWCLK	76	PA14_JTCK/SWCLK
ADS1235_nCS	77	PA15_JTDI

ULPI_D1	34	PB0
ULPI_D2	35	PB1
QSPI_P1_CLK	36	PB2
SW0	89	PB3_JTDO/SW0
SPARE_PB4	90	PB4_JTRST
ULPI_D7	91	PB5
QSPI_P1_nCS	92	PB6
SPARE_PB7	93	PB7
EEPROM_SCL	95	PB8
EEPROM_SDA	96	PB9
ULPI_D3	46	PB10
ULPI_D4	47	PB11
ULPI_D5	51	PB12
ULPI_D6	52	PB13
CONFIG_SPI_MISO	53	PB14
CONFIG_SPI_MOSI	54	PB15

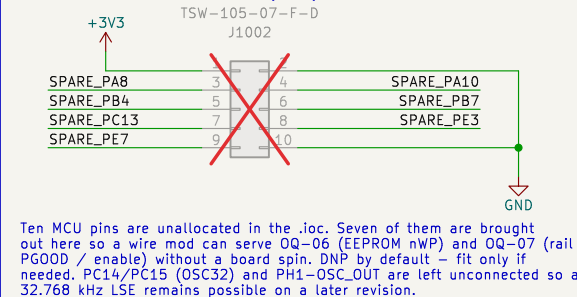
ULPI_STP	15	PC0
V24_MON	16	PC1
ULPI_DIR	17	PC2_C
ULPI_NXT	18	PC3_C
MOTOR_I_B	32	PC4
VBUS_MON	33	PC5
MOTOR_ENCODER_A	63	PC6
MOTOR_ENCODER_B	64	PC7
MOTOR_ENCODER_I	65	PC8
ADS1235_CLKIN	66	PC9
ADS1235_SCLK	78	PC10
ADS1235_MISO	79	PC11
ADS1235_START	80	PC12
SPARE_PC13	7	PC13
PC14-OSC32_IN	8	PC14
PC15-OSC32_OUT	9	PC15

ADS1235_nRESET	81	PD0
ADS1235_nRDY	82	PD1
LIM_A	83	PD2
ADS1120_nCS	84	PD3
LIM_B	85	PD4
DRV8323_CAL	86	PD5
ADS1235_MOSI	87	PD6
DRV8323_nCS	88	PD7
DBG_TX	55	PD8
DBG_RX	56	PD9
DRV8323_EN	57	PD10
QSPI_P1_I00	58	PD11
QSPI_P1_I01	59	PD12
QSPI_P1_I03	60	PD13
HALL2	61	PD14
HALL3	62	PD15

MCU PORTS (cont.)

BTN_1	97	PE0
BTN_2	98	PE1
QSPI_P1_I02	1	PE2
SPARE_PE3	2	PE3
USB3320_nRESET	3	PE4
SYNC_TRIG	4	PE5
LIMIT_nBRK	5	PE6
SPARE_PE7	37	PE7
MOTOR_PWM_AL	38	PE8
MOTOR_PWM_AH	39	PE9
MOTOR_PWM_BL	40	PE10
MOTOR_PWM_BH	43	PE11
MOTOR_PWM_CL	42	PE12
MOTOR_PWM_CH	43	PE13
HALL1	44	PE14
DRV8323_nFAULT	45	PE15

SPARE MCU I/O BREAKOUT (DNP)



SHEET NOTES

STM32H723VET6 + USB3320 ULPI PHY + USB-C (data only) + OCTOSPI1 PSRAM
Pin map authority: hardware/cubemx/ARIA_SRB_FAFF_2_POC_CBs_1.ioc (DEC-0013) – nothing here re-pins it.
Local labels stay on this sheet: ULPI_*, QSPI_P1_*, USB_*, HSE_CLK_24M, USB3320_nRESET, SPARE_*, +3V3_USB, +1V8_USB.
Hierarchical labels leave it; the full list, and the naming choices behind it, are in docs/decisions/actuator-sch-mcu.md.
PC2_C / PC3_C carry ULPI_DIR and ULPI_NXT; firmware MUST close the SYSCFG_PMC analog switch (PC250/PC350) before the ULPI link will run – the pins are otherwise isolated from the digital I/O. DS13313 6.3.16 Table 56.

TEST ACCESS (TEST_PLAN 4). TPs: NRST (hook), BOOT0, HSE clock (dual TP at the oscillator, block B), +3V3_USB and +1V8_USB (block E). MC02 leaves the sheet as ADS1235_CLKIN and is probed in loadcellLafe, which owns that net's break.
Supply-pin groups are reachable at the decoupling banks in blocks A, D, E and F. DELIBERATELY NO test points on ULPI or OCTOSPI (DEC-0018 ruling 1 / TEST_PLAN 2.2).

ROOT SHEET NEEDS. Every hierarchical label on this sheet is a sheet pin on the mcu block of the root sheet, drawn and wired by the integration pass that closed DEC-0009; project ERC is clean at severity-all with none of them suppressed. The list and the direction of every net is in docs/decisions/actuator-sch-mcu.md, and the root-sheet geometry in docs/decisions/actuator-sch-integrate.md.

Reasoning: docs/decisions/actuator-sch-mcu.md		
No test points on the ULPI or OCTOSPI buses (DEC-0018)		
USB-C is DATA ONLY – no USB-PD, no bus power (DEC-0002)		
Pin map authority: hardware/cubemx (.ioc) – DEC-0013		
Amodo Design		
Sheet: /mcu/		
File: mcu.kicad_sch		
Title: CBs_1 – MCU (STM32H723VET6) + USB + QSPI		
Size: A3	Date: 2026-09-02	Rev: 0.2
KiCad E.D.A. 9.0.8		Id: 10/11

R1101
0R

R1102
0R

+24V

V24_MOT

TP1101

TP1102

R101 isolation link, R102 current-measurement break.
Both fitted. Opening R101 leaves the whole logic side
powered with the actuator dead – TEST_PLAN.md 3.3, step 8.

C1101
100uF

C1102
100uF

C1103
10uF

C1104
100nF

C1105
10nF

VBUS_MON

R1103
100k

TP1103

R1104
12k

Bulk store 210 uF; DC-link ripple < 0.15 V at 1.9 A, 20 kHz.
VBUS_MON 100k/12k: 3.30 V full scale = 30.8 V bus; 24.0 V
reads 2.571 V. PC5 / ADC2_INPB (DEC=0011, OQ=01).

RT1101 sits on the power stage, between the low-side FETs. 10k NTC, $B = 3435\text{ K}$, 10k pull-up: 1.65 V at 25 C, 0.86 V at 85 C. PA4 / ADC1_INP18 (DEC-0011).

Encoder supply – FIT ONE ONLY.
R1106 fitted = 3V3 (CALC budget).
R1107 = 5 V read-head option.
Pull-ups always go to 3V3, so the
MCU never sees 5 V from an open-
collector encoder or hall output.

The diagram shows two 6-pin connectors, J1101 and J1102, connected to a microcontroller's I/O pins. J1101 is connected to VENC, TP1105, MOTOR_ENCODER_AD, MOTOR_ENCODER_BD, and MOTOR_ENCODER_ID. J1102 is connected to VENC, HALL1D, HALL2D, and HALL3D. Pull-up resistors R1108, R1109, R1110, R1111, R1112, and R1113 are connected to the output pins. Capacitors C1107 and C1108 are connected to the VENC pin. The VENC pin is also connected to a 5V supply through a resistor R1106. The TP1105 pin is connected to a 5V supply through a resistor R1107. The HALL1D, HALL2D, and HALL3D pins are connected to a 5V supply through resistors R1111, R1112, and R1113 respectively. The MOTOR_ENCODER_AD, MOTOR_ENCODER_BD, and MOTOR_ENCODER_ID pins are connected to a 5V supply through resistors R1108, R1109, and R1110 respectively. The VENC pin is also connected to a 5V supply through a resistor R1106. The TP1105 pin is connected to a 5V supply through a resistor R1107. The HALL1D, HALL2D, and HALL3D pins are connected to a 5V supply through resistors R1111, R1112, and R1113 respectively. The MOTOR_ENCODER_AD, MOTOR_ENCODER_BD, and MOTOR_ENCODER_ID pins are connected to a 5V supply through resistors R1108, R1109, and R1110 respectively.

J1101

1 VENC

2

3

4

5

6

1x6 2.54mm

TP1105

R1106 0R

R1107 0R

+5V

C1107 10uF

C1108 100nF

R1108 2k2

MOTOR_ENCODER_AD

R1109 2k2

MOTOR_ENCODER_BD

R1110 2k2

MOTOR_ENCODER_ID

J1102

1 VENC

2

3

4

5

6

1x6 2.54mm

R1111 10k

HALL1D

R1112 10k

HALL2D

R1113 10k

HALL3D

J102 A/B/Z -> TIM3 CH1/CH2/CH3 (PC6/PC7/PC8). J103 halls -> PE14/PD14/PD15 (GPIO). Motor not fixed (DEC-0004): this pinout suits every candidate BLDC in the design log.

R1114 isolates the gate-driver supply from the motor bus. VDRAIN comes from the high-side FET drains, not from VM, so the VDS monitor sees the real bridge rail (SLVSDJ3D 11.1).

TIM1 CH1/CH1N..CH3/CH3N (PE8-PE13), 6x PWM mode, dead time in the MCU. INHx/INLx have internal 100k pull-downs; ENABLE low forces GHx to SHx and GLx to PGND, so the bridge is off in MCU reset.

VREF ties the CSA reference to the same +3V3 the STM32 ADC uses, so phase current is ratiometric. DVDD is the DRV internal 3.3 V regulator: decoupling only, no load.

SAFETY—CRITICAL BREAK PATH. R1119 is FITTED by default and marked on the silkscreen. DRV8323_nFAULT (open drain) drives TIM1_BKIN on PE15; opening R1119 exercises the LIMIT_nBRK path alone (TEST_PLAN.md 3.3, bring-up step 8). R1120 holds PE15 high while R1119 is out. R1116/R1117 are the SPI2 stub links – remove both to lift U1101 off the shared bus; SDD0 is open drain and HI-Z while nSCS is high, so MISO needs no link. R1121 keeps ENABLE – and the bridge – off until PD10 drives high.

PEAK CURRENT LIMIT (REQ-SF-03, closes QQ-08). One shunt value serves both variants; the limit is an SPI setting plus a firmware threshold, not a board change. QR15 with CSA gain 5 V/V gives +/-1.87 A full scale on V50N, so the 75 N (1.843 A) limit lands on the last ADC code; gain 20 V/V gives +/-0.467 A on V10N, 15 N at 79 % of scale. SEN_LVL_00b trips the hardware comparator at 1.667 A = 67.8 N, below the V50N cell limit. A V10N build wanting that trip at 15 N fits OR68.

POWER STAGE. No series gate resistors and no G-S pull-downs – slew rate is set by IDRIVE/IDRIVE over SPI, and external gate parts degrade the VDS and gate-fault monitors. SPx is the VDS monitor return as well as the CSA input, so keep it Kelvin to the FET source. LAYOUT: shunt -> PGND -> bulk-cap negative is ONE tight loop, joined to system ground at a single point at the bulk capacitors.

Root sheet wired by the integration pass (DEC-0009)	
Requirements: docs/REQUIREMENTS.md Tests: docs/TEST_PLAN.md	
Reasoning: docs/decisions/actuator-sch-motor.md	
3-phase inverter, phase-current sense, motor and rotary-encoder interface	
Amodo Design	
Sheet: /motor_drive/	
File: motor_drive.kicad_sch	
Title: CBs_1 – Motor Drive (DRV8323S)	
Size: A3	Date: 2026-09-03
KiCad E.D.A. 9.0.8	Rev: 0 Id: 11